

- 7 (a) (i) 1 By Newton's 3rd law, the rocket exerts a force on the gases so the gases exert an equal and opposite force on the rocket. B1
- By Newton's 2nd law, this (net) force on the rocket will cause it to accelerate. B1
- 2 Total momentum of rocket and gas as a system remains constant since there is no (net) external force acting on it. B1
- B1

After engine is turned on, gases gain momentum to the left, rocket will gain equal magnitude of momentum to the right.

$$3 \text{ Force on gases} = \text{rate of change of momentum of gases} \\ = Rv$$

B1

$$\text{So force on rocket} = Rv$$

B1

$$\text{Hence } ma = Rv$$

B1

$$\Rightarrow a = \frac{Rv}{m}$$

A0

(ii) After the fuel in the 1st stage is used up, the acceleration of the rocket will be higher since m decreases. M1

Hence, the 2 stage rocket will have a larger final speed. A1

(b) (i) 1 constant net force = weight of the girl = 350 N

C1

$$\text{mass of the girl} = \frac{350}{9.81}$$

$$= 36 \text{ kg}$$

A0

2 Time taken for the girl to fall before touching the trampoline = 0.50 s

$$v = u + at$$

$$= 0 + 9.81(0.50)$$

$$= 4.91 \text{ m s}^{-1}$$

A1

3 Maximum upward net force, $F_{net} = 1400 \text{ N}$

$$F_{net} = N - W$$

C1

$$N = F_{net} + W = 1400 + 350$$

$$= 1750 \text{ N}$$

A1

(ii) 1 The net force decreases as normal contact force increases. B1

Therefore, the acceleration decreases.

Speed increases at a decreasing rate. B1

At point D, the net force is zero, normal contact force equals weight and the girl has maximum speed. B1

2 change in momentum = area under graph from C to D

$$= \frac{1}{2} \times 350 \times (0.53 - 0.50)$$

C1

$$= 5.3 \text{ Ns}$$

A1

3 $m(v_D - v_C) = \Delta p$ e.c.f. from part (ii)2

C1

$$36(v_D - 4.9) = 5.0$$

$$v_D = 5.0 \text{ m s}^{-1}$$

A1

